**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. A variable storing a running total.

Q2. Two keys map to the same initial bucket.

Q3. A reference to the first node.

Q4. A stack: the most recent operation must be undone first.

### You Do pages · Independent questions

Q1. Queue (FIFO).

Q2. Hash table: hashing IDs supports expected fast exact-key retrieval; stacks expose only the latest item.

Q3. Balanced BST allows ordered traversal and O(log n) search; a hash table has expected O(1) lookup but does not inherently keep keys ordered.

### Exit Ticket · Model answer

False. Collisions can make lookup much slower, sometimes O(n).

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 206 · Data structure selection challenge

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is an accumulator?

Q2. What is a collision?

Q3. What is a head pointer?

Q4. Which structure is used for Undo and why?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

LIFO — Last In, First Out.

FIFO — First In, First Out.

lookup — Retrieve a value using a key.

balanced BST — Tree kept at relatively small height.

trade-off — Benefit balanced against a drawback.

### New learning

Identify the operation — The print job that arrived first must leave first.

Match the structure — Enqueue adds jobs; dequeue removes the oldest one. This is a queue.

Justify and qualify — A stack gives LIFO, which is the wrong order. Other systems may also need priority queues if priority overrides arrival order.

### Worked example

A browser Back button must return to the most recently visited page. Choose a structure.

1. Predict — The last visited page is the first one that must be removed.

2. Explain — Use a stack: push visits and pop the most recent page.

## We Do · Practise with guidance

A ticketing desk must serve requests in arrival order. Which structure?

```
Tickets arrive: A → B → C. Serve one ticket.
```

Predict

The first request A must be the first one handled.

Explain

Use a queue, because FIFO removes the oldest item.





BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Select a structure for processing jobs in arrival order.

### Question 2

Choose stack or hash table to retrieve customer records by unique ID. Justify.

### Question 3

Compare a balanced BST and hash table for lookup and sorted output.

## Exit Ticket · Answer before checking

A hash table is always O(1) even in the worst case.